

Q. No. 2 (a)

What do you mean by the term drift velocity and mobility of an electron? Derive the expression for mean thermal velocity of electron: $v_{th} = (3kT/m)^{1/2}$

Ans

A drift velocity is the average velocity attained by charged particles, such as electrons, in a material due to an electric field.

The mobility of an electron is the measure of how fast an electron moves through a metal or semiconductor under the influence of external field.

Thermal velocity is the velocity that a particle in a system would have if its kinetic energy were equal to the average energy of all the particles of the system.

In equilibrium there is a distribution of velocities among all of the particles. Some move "fast" others move "slowly". That means that there is a distribution of kinetic energy. Call it E_{avg} by adding up the energies of all of the particles and then divide that by the no. of particles. It turns out that the average kinetic energy can be calculated for systems in equilibrium. For an ideal gas, $E_{avg} = \frac{3}{2} kT$.

Average energy is proportional to the temp

of the system. There are similar expressions for more complicated systems, but the average energy is always proportional to the thermal

The thermal velocity is the speed that a particle would have if its K.E happened to be equal to the average value. Thus for an ideal gas

$$\frac{1}{2}mv^2 = \frac{3}{2}KT$$

$$\text{and } v = \sqrt{\frac{3KT}{m}} \quad \text{or, } v_{th} = (3KT/m)^{1/2}$$

For a system of particles of known mass, the thermal velocity depends only on the temperature and a universal constant. Thus, it is an indirect measure of temp.

Q. Define and explain the effective mass of electron with in a crystal. How do you understand negative and infinite effective mass of electron?

or

The effective mass is a quantity that is used to simplify band structures by modeling the behaviour of a free particle with that mass.

For

Effective mass of electron

When an electron in a periodic potential accelerated by an electric field or a magnetic field then the mass of the electron varies with velocity. This means the mass of the electron is function of velocity and is termed as effective mass of the electron (m^*).

Force acting on the electron $F = m^* a$

$$a = \frac{F}{m^*} = -\frac{q}{m^*} E$$

According to quantum theory an electron moving with velocity v can be treated as wave packet moving with group velocity

$$v_g = \frac{d\omega}{dk}$$

$$v_g = \frac{d(2\pi \hbar^{-1} \hbar v)}{dk} \quad (\because \omega = 2\pi \hbar^{-1} \hbar v)$$

$$= 2\pi \hbar^{-1} \hbar \frac{dv}{dk}$$

$$v_g = \frac{\partial \bar{n}}{\partial k} \frac{dE}{h} \quad (\because E = h\nu \rightarrow \nu = \frac{E}{h})$$

$$v_g = \frac{\partial \bar{n}}{\partial k} \frac{dE}{h}$$

$$v_g = \frac{1}{h} \frac{dE}{dk} \rightarrow \textcircled{1} \left[\because h = \frac{h}{2\pi} \right]$$

Find acceleration $a = \frac{dv_g}{dt}$

Diff eqn ① w.r.t t

$$a = \frac{dv_g}{dt} = \frac{d}{dt} \left[\frac{1}{h} \frac{dE}{dk} \right]$$

$$a = \frac{1}{h} \frac{d}{dt} \left[\frac{dE}{dk} \right] \frac{dk}{dt}$$

$$a = \frac{1}{h} \frac{dk}{dt} \frac{d^2 E}{dk^2}$$

$$a = \frac{1}{h} \frac{dk}{dt} \frac{d^2 E}{dk^2} \rightarrow \textcircled{2}$$

Find $\frac{dk}{dt}$ we know $p = \hbar k \rightarrow \textcircled{3}$

Diff eqn ③ w.r.t t

$$\frac{dp}{dt} = \hbar \frac{dk}{dt}$$

$$F = \hbar \frac{dk}{dt}$$

$$\therefore \frac{dk}{dt} = \frac{F}{\hbar} \rightarrow \textcircled{4}$$

Put eqn ④ in eqn ②

$$a = \frac{1}{h} \frac{F}{\hbar} \frac{d^2 E}{dk^2}$$

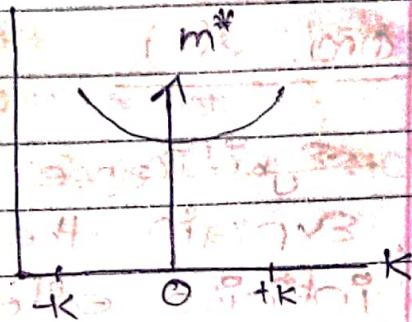
$$a = \frac{F}{\hbar^2} \frac{d^2 E}{dk^2} \rightarrow \textcircled{5}$$

Comparing eqn (6) & (1)

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2}$$

$$m^* = \hbar^2 \left[\frac{d^2 E}{dk^2} \right]^{-1}$$

$$m^* = \hbar^2 \frac{d^2 E}{dk^2}$$



The above eqn gives effective mass of electron.

If the external force is still positive, which means that the electron acceleration now becomes negative, i.e. the electron decelerates. The negative effective mass tells you that the electron responds to the field opposite to how a free electron would.

2012-Spring

(d) 1.0001

Q.11 prove that the thermal velocity of electron at equilibrium is $v_{th} = [3kT/m]^{1/2}$

Sol: -

Applying kinetic molecular theory of gases to gas molecules contained in a container we can write the final theoretical derivation in the form

$$PV = \frac{Nm\bar{v}^2}{3}$$

where, p = pressure of gas inside the container
 V = volume of the container

N = number of gas molecules in container

m = mass of the gas molecules.

$\bar{v}^2$ = mean square velocity of molecules.

The experimental observation is

$$PV = \frac{N}{N_A} RT = nRT$$

where, N_A is Avogadro's no.

R = universal gas constant

T = temperature.

$n = N/N_A$ = number of moles of gas

For N molecules gas, every molecule will have its random velocity. The change in momentum of the molecule following collision with the wall of the container is $\Delta p_1 = 2mv_x$ where m is the molecular mass. Time interval between collision will be $\Delta t = 2a/v_x$ where a is the side of container. Force exerted by this molecule on the face A of the

Container $P \Delta t = \frac{\Delta p}{\Delta t} = \frac{m v_x^2}{a}$

And the total pressure exerted by N molecules on face A due to total force is $p = \frac{\text{Total Force}}{\text{Area}}$

$$= \frac{m}{a^3} (v_{x1}^2 + v_{x2}^2 + \dots + v_{xN}^2)$$

$$p = \frac{mN}{V} v_x^2$$

where $N v_x^2 = v_{x1}^2 + v_{x2}^2 + \dots + v_{xN}^2$

Since the molecules are in random motion & collide randomly with each other, mean square velocity in x direction will be equal to that in y & z directions. so, $v_x^2 = v_y^2 = v_z^2$

The velocity of any molecule is given by

$$v^2 = v_x^2 + v_y^2 + v_z^2 = 3v_x^2$$

v^2 = mean square velocity of the molecule.

We will find that average kinetic energy per degree of freedom is (can be derived) $KT/2$. So for three degrees of freedom this is $3KT/2$. Average K.E per mole is $mv^2/2$.

$$\frac{K.E}{2} = \frac{mv^2}{2} = \frac{3}{2}KT$$

$$v_{th} = \left(\frac{3KT}{m} \right)^{1/2}$$

where v_{th} is the mean thermal velocity of electrons.

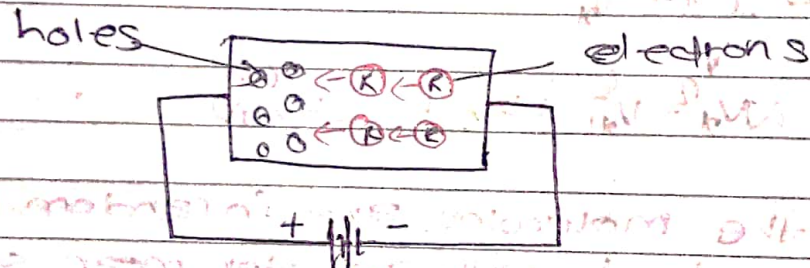
2018-Fall

Q. 1. A semiconductor has a band gap of 1.1 eV. Calculate the intrinsic carrier concentration at 300 K.

Q. 2. (a)

Derive the expression that relates mobility and conductivity.

(a) The electron mobility characterises how quickly an electron move through a metal or semiconductor.



When an electric field E is applied across a piece of material, the electrons respond by moving with an average velocity called the drift velocity, V_d . Then the electron mobility μ is defined as:

$$V_d \propto E$$

$$V_d = \mu E$$

$$\Rightarrow \mu = \frac{V_d}{E}$$

μ depends upon (i) Temperature
(ii) material
(iii) motion of electron or holes.

Conduction Band - electrons

Valence band - holes

Semiconductor.

Electrons & holes acceleration.

$$e_n = \frac{V_n}{E}$$

$$e_p = \frac{V_p}{E}$$

Consider a semiconductor having an electron density in the conduction band is n per unit volume.

The charge which is available for conduction per unit volume is ne . If this charge will move with a velocity v_n in the field E , then the current density is given by

$$J_n = nev_n$$

Similarly for holes

$$J_p = p e v_p$$

Now conductivity

$$\sigma_n = \frac{J_n}{E} = \frac{nev_n}{E} \quad \text{--- (A) (due to electrons)}$$

$$\sigma_p = \frac{J_p}{E} = \frac{p e v_p}{E} \quad \text{--- (B) (due to holes)}$$

Total conductivity due to electrons & holes

$$\sigma = \sigma_n + \sigma_p = \frac{nev_n}{E} + \frac{p e v_p}{E}$$

$$\sigma_p = \frac{J_p}{E} = \frac{p e v_p}{E}$$

$$\sigma = nev_n + p e v_p$$

2014 - Fall

George E. 108

Qno. 2(a)

101.01

Significance of effective mass of solid electronic.

Sol: -

The effective mass is used in transport calculations, such as transport electrons under the influence of fields or carrier gradients, but it is also used to calculate the carrier density and density of states in semiconductors.

In applying the usual equations of electrodynamics to charge carriers in a solid, we must use altered values of particle mass. we named it effective mass.

Qno. 2(b)

Calculate the Fermi energy at 0K (Kelvin) for copper given its density is 8.96 g/cm^3 & atomic mass 63.5 g/mol.

Sol: -

Assuming each atom contributes one electron to the conduction band, the electron concentration of copper is

$$n = \frac{\rho N_A}{M_{at}} = \frac{(8.96 \text{ g/cm}^3)(6.022 \times 10^{23} \text{ /mol})}{(63.5 \text{ g/mol})}$$

$$= 8.5 \times 10^{22} \text{ cm}^{-3}$$

The fermi energy at 0K is given by

$$E_{F0} = (h^2/8m_e) (3n/2\pi)^{2/3}$$

$$E_{F0} = \frac{(6.624 \times 10^{-34} \text{ Js})^2}{8 \times 9.1 \times 10^{-31} \text{ Kg}} \left(\frac{3 \times 8.5 \times 10^{18} \text{ m}^{-3}}{2\pi} \right)^{2/3}$$

$$= 1.1 \times 10^{-18} \text{ J} = 7 \text{ eV}$$

2014 - Spring

Qno. 2(a)

Show that the conductivity of electrons within metallic conductor is product of charge density and mobility.

Ans

2015-Spring

Qno. 7 (c)

FCC crystal structure

Here the atoms are placed in the case of SC at all the corners of the unit cell & additional atoms are placed at the intersection of face diagonals in all faces. Half of the atom at each of the face diagonals belongs to the unit cell. So, the total number of atoms in a unit cell equals to $(1 + 6 \times \frac{1}{2} = 4)$ four. Each FCC unit cell has four atoms in it. The number of neighbouring atoms is termed as Coordination no. In FCC structure the coordination no. is 12.

2016-Spring

Qno. 8 (a)

Electron in undoped gallium arsenide have a mobility of $8.8 \text{ cm}^2/\text{V.s}$. Calculate average time between collision. Calculate distance travel between two collision. The average velocity of electron is 10^7 m/s .

2017-spring

phys 2-210

bro. 2(a)

(1) F. 1000

obtain expression to μ

State Fick's law of diffusion. Derive Einstein's relationship between mobility and diffusion coefficient.

By

Fick's law of diffusion describes how particles under a random thermal motion tend to spread from a region of higher concentration to a region of lower concentration.

The mobility ' μ ' expresses the ability of carriers to drift while the diffusion.

It gives the direct relation between diffusion coefficients and mobility of charge carrier.

At equilibrium condition, drift current balances and opposite to the diffusion current.

$$J_D = -D \frac{\partial n}{\partial x}$$

$\therefore \text{Drift} = -\text{Diffusion}$

$$ne \mu E = -eD \frac{\partial n}{\partial x} \quad \text{--- (1)}$$

Einstein compared the movement of charge carriers with the gas molecules in a container.

Acc. Boltzmann's statistics, the concentration of gas molecules can be written as.

$$n = C \exp \left[\frac{-eE_x}{k_B T} \right]$$

Diff above eqn w.r.t x partially:

$$\frac{\partial n}{\partial x} = C \cdot \exp \left[\frac{-eE_x}{k_B T} \right] \left(\frac{-eE}{k_B T} \right)$$

$$\frac{\partial n}{\partial x} = n \left(\frac{-eE}{k_B T} \right) \quad \text{--- (2)}$$

put eqn (2) in eqn (1) we get

$$n e E = - e D_e n \left(\frac{-eE}{k_B T} \right)$$

$$\mu_e = \left(\frac{e D_e}{k_B T} \right)$$

$$\frac{D_e}{\mu_e} = \frac{k_B T}{e} \quad (\text{for } e^-)$$

$$\text{similarly, } \frac{D_h}{\mu_h} = \frac{k_B T}{e} \quad (\text{Exp is for holes})$$

$$\frac{D_e}{D_h} = \frac{\frac{k_B T \mu_e}{e}}{\frac{k_B T \mu_h}{e}}$$

$$\boxed{\frac{D_e}{D_h} = \frac{\mu_e}{\mu_h}}$$